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A multi-force microscale model for lunar dust detachment and dispersion induced by rover mobility

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Abstract Lunar rover mobility inevitably induces dust lifting, posing risks to onboard instruments. This study develops a physics-based model that incorporates van der Waals forces, electrostatic interactions, and Coulomb forces to analyze the detachment and dispersion behavior of lunar dust. Numerical simulations using MATLAB are conducted to determine the critical detachment radius under varying driving speeds and slip rates. Results indicate that increasing either speed or slip rate reduces the detachment threshold from 19.11 to 17.79 μm. Further analysis of the simulated dust-dispersion images reveals a clear stratification in ejection height as a function of particle size. Under high-velocity and high-slip-rate conditions, small particles (radius < 20 μm) dominate the upper portion of the dust plume, reaching vertical propagation heights of up to 1.8 m, which substantially exceed the height of the rover itself. Validation against Apollo mission imagery confirms the model's applicability. These findings provide theoretical guidance for dust mitigation strategies in future lunar rover missions.

Keywords lunar dust, fender, detachment, smooth rotating wheel, theoretical model

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1 Introduction

As China's lunar exploration program surpasses its 20th anniversary, the nation has successfully achieved the three-step strategic goal of "orbiting, landing, and returning", making significant contributions to human lunar exploration [1–3]. The planning and execution of follow-up missions, such as Chang'e 6, indicate that China's lunar program has entered a more advanced phase, involving exploration of the Moon's south pole and the construction of a lunar base. The Moon exhibits substantial environmental differences from Earth, and its surface has developed a layer of extremely fine, powdery particles-lunar dust-formed through long-term meteorite impacts and solar radiation. Lunar dust particles are small, hard, angular, and electrically charged. The movement of manned lunar rovers causes extensive dust-related issues, posing a significant challenge to lunar exploration [4]. Lunar dust not only causes mechanical wear to equipment but also undermines the stability of thermal control systems, leads to failures in sealing mechanisms, and reduces the sensitivity of optical systems [5]. Therefore, accurately predicting and analyzing the trajectory of lunar dust lifted by rover activity is of critical importance.

Due to the electrically charged nature of lunar dust and the unique characteristics of the lunar surface environment, the contact force between lunar dust particles and rover wheels differs from that on Earth. Numerous studies have investigated the adhesion behavior of lunar dust with respect to electrostatic force, van der Waals force, vacuum conditions, and surface material variation, as well as the relationship between adhesion force, particle size, and contact state [6–11]. In addition, lunar dust is subject to electrostatic levitation. In 2006, Stubbs et al. [12] proposed a dynamic fountain model of lunar dust and derived formulas for the maximum levitation height of dust particles (Z_{max}) and the maximum radius of levitated particles (R_{max}). On the sunlit side of the Moon, solar radiation and the solar wind induce photoelectron emission, causing lunar dust particles to become posi-

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tively charged [13]. These positively charged particles are more susceptible to upward-directed electric forces within the near-surface lunar electric field, making them more likely to detach and become suspended when subjected to external disturbances.

Although the aforementioned studies have advanced the understanding of adhesion modeling and particle behavior, investigations into the dust-lifting mechanisms associated with lunar rover wheels remain relatively limited to date. In 1970, the Marshall Space Flight Center in Alabama, USA, conducted an experiment on wheel-induced lunar dust. The study found that baffles could effectively block raised lunar dust particles and that low gravity prolonged the airborne duration of these particles. However, vacuum conditions had a more significant influence than gravity, leading to more rapid settling of lunar dust particles under vacuum [14]. As traditional laboratory conditions cannot replicate both the vacuum and low-gravity characteristics of the lunar environment, most current research relies on numerical simulations. Some studies have applied finite element and discrete element methods to investigate dust lifting by lunar rover wheels, analyzing wheel-soil interactions under low-gravity conditions [15–19], dust lifting behavior across various wheel tooth structures [20], and the dust suppression performance of different baffle configurations [4]. However, existing studies have predominantly focused on macroscopic simulations of soil-wheel interactions, while systematic descriptions of the microscale contact-separation mechanisms between lunar dust particles and rover wheels remain limited. In particular, the particle-size thresholds governing lunar dust resuspension are not yet well characterized, and most prior work has emphasized negatively charged particles, overlooking the dust-lifting behavior driven by positively charged grains on the sunlit surface. In addition, the conventional finite element method (FEM), which relies on continuum assumptions, is unable to resolve the nanoscale adhesive and electrostatic forces that dominate lunar dust detachment. Discrete element method (DEM) models, although capable of representing granular interactions, require extensive parameter calibration for lunar regolith and become computationally prohibitive at the micron scale. As a result, DEM simulations typically cannot capture particle behavior below ~100 μm, limiting their ability to reproduce real lunar dust lofting phenomena. These limitations highlight the need for a mechanism-based framework that explicitly incorporates the governing microscale forces to better model lunar dust detachment and dispersion. In addition, due to the inherent challenges in reproducing lunar environmental conditions experimentally and the scarcity of in situ data, current simulation results remain difficult to validate against actual dust-lifting behavior, leading to notable gaps in experimental verification.

To further investigate the mechanism of dust lifting caused by lunar rover wheels, this study establishes a multi-force coupling model that incorporates adhesion, electrostatic, and Coulomb forces acting on positively charged lunar dust particles. A MATLAB-based simulation framework is developed to calculate the critical particle sizes for detachment under varying travel speeds and slip rates. The resulting dust trajectories, accounting for the influence of baffle structures, are compared with imagery from the Apollo missions to evaluate the model's applicability under sunlit lunar surface conditions.

The key contributions of this work are as follows.

A multi-force coupling model incorporating adhesion, electrostatic, and Coulomb forces to predict the critical conditions for lunar dust lifting under varying speeds and slip rates;

A numerical simulation framework using MATLAB to calculate critical particle sizes and simulate dust trajectories induced by lunar rover wheels (the overall modeling framework is illustrated in Section 2.5);

Visual verification of simulated dust behavior against Apollo rover imagery, supporting the model's accuracy under sunlit lunar surface conditions.

2 Methodology

Before detailing the force models, it is important to clarify the applicability of the proposed framework with respect to particle size. The present study focuses on lunar dust particles within the radius range of 18–45 μm, in which adhesion, electrostatic interactions, and centrifugal forces jointly dominate the detachment behavior. Particles significantly smaller than this range tend to remain strongly adhered to the wheel surface, while larger particles (> 45 μm) exhibit inertia-dominated behavior that falls outside the scope of the current microscale force-based model. The detailed size-dependent analysis is presented in Section 2.4.2.

2.1 Contact force between lunar dust particles and wheels

2.1.1 Van der Waals forces

Van der Waals forces play a major role in the adhesion of lunar dust particles to contact surfaces. In classical contact mechanics theory, both the Johnson-Kendall-Roberts (JKR) model [21] and Derjaguin-Muller-Toporov (DMT) model [22] can be used to estimate van der Waals forces between spherical particles and surfaces based on surface energy and particle size. However, these models are only applicable to smooth particles and surfaces, and each is suitable for a distinct application range. As the JKR model is widely used in adhesion calculations, it is adopted in this study for modeling purposes:

$$\gamma = (\gamma_1\gamma_2)^{1/2}, \tag{1}$$

$$E^* = \left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2}\right)^{-1}, \tag{2}$$

$$a_{\mathrm{JKR}} = \left(\frac{9\pi R^2\gamma}{E^*}\right)^{1/3}, \tag{3}$$

$$F_v^{\mathrm{JKR}} = \frac{E^* a_{\mathrm{JKR}}{}^3}{3R}, \tag{4}$$

where $\gamma_1 = 0.06$ J/m^2 is the surface energy of the lunar dust [4], a representative value of $\gamma_2 = 0.9$ J/m^2 was adopted based on the commonly reported surface energy range of engineering polymer and rubber materials and γ is the effective surface energy; R is the radius of the lunar dust particles; E_1 = 65 MPa and E_2 = 70 MPa are the Young's modulus of the lunar dust and wheel [4], respectively; the Poisson's ratio of the lunar dust particles [4] $\mu_1 = 0.33$; the Poisson's ratio of the wheel material [4] $\mu_2 = 0.2$; E^* is the effective Young's modulus; a_{JKR} is the radius of the contact circle resulting from JKR's theoretical calculation of particulate matter in contact with the contact surface; F_v^{JKR} is the adhesion force of particles.

In real environments, both lunar dust particles and the wheel surface exhibit surface roughness. Surface roughness reduces the actual contact area between the particles and the contact surface, thereby significantly decreasing the effective van der Waals force. Assuming that C_r is the ratio of the actual contact area to the theoretical contact area between lunar dust particles and the contact surface, You and Wan [23] derived a formula for calculating C_r by analyzing the roughness effects on particle-surface interactions, as shown in Figure 1. The highly irregular morphology and sub-100-μm size of lunar regolith grains make it impractical to directly measure the surface roughness of individual particles. Because lunar regolith is predominantly composed of silica-rich basaltic and glassy materials, experimentally measured roughness values of SiO_2 particles are commonly used as surrogate parameters in lunar dust adhesion studies. Accordingly, the present work adopts previously reported roughness measurements of SiO_2 particles [24], whose angularity and micro-texture closely resemble those of lunar dust grains. According to the typical roughness values of machined aluminum-alloy surfaces reported in the Mechanical Design Handbook (6th ed, Mechanical Industry Press) [25], the surface roughness of the rover wheel is taken as 1 μm, which falls within the standard range of 0.4–3.2 μm for rolled or machined aluminum-alloy surfaces.

As described by eqs. (5)–(8), the contact-area ratio Cr depends on the effective surface roughness σ and the deformation-related displacements δ1 and δ2. σ represents the combined roughness of the particle and wheel surfaces, while δ1 and δ2 describe the squeezing and stretching deformation of the contacting asperities, respectively. A larger effective roughness generally results in a smaller Cr, indicating a reduced effective contact area and, consequently, weaker van der Waals adhesion.

$$C_r = \frac{1}{2}\left[\mathrm{erf}\left(\frac{\delta_1+\delta_2-h_{\min}}{\sqrt{2\sigma}}\right)+1\right], \tag{5}$$

$$\sigma = \sqrt{\left(\sigma_1^2+\sigma_2^2\right)}, \tag{6}$$

$$\delta_1 = 3.53\left(\frac{\sigma^{1/2}\gamma}{E^*}\right)^{2/3}, \tag{7}$$

$$\delta_2 = 2.914\left(\frac{\sigma^{1/2}\gamma}{E^*}\right)^{2/3}, \tag{8}$$

where the roughness of the lunar dust particles $\sigma_1 = 1.5$ nm; the roughness of the wheel surface $\sigma_2 = 1$ μm; σ is the effective roughness; the two rough surfaces in contact with the smallest distance $h_{\min} = 2.7\sigma$; δ_1 is the squeezing distance due to deformation of the particulate matter in contact due to van der Waals forces; δ_2 is the stretching distance due to deformation of the particulate matter in contact due to van der Waals forces.

From this, the formula for the van der Waals force between the moon dust particles and the wheel surface can be obtained by considering the effect of roughness:

$$F_v = \frac{E^* C_r^{3/2} a_{\mathrm{JKR}}^3}{3R}. \tag{9}$$

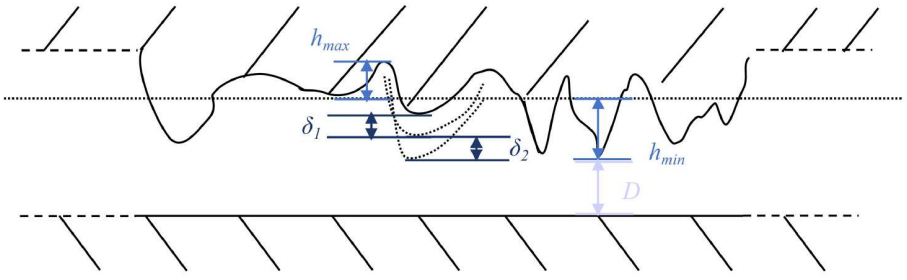


Figure 1 (Color online) Schematic illustration of surface roughness parameters δ_1, δ_2, $h_{\min}$, and the effective contact area used in the calculation of the roughness correction factor C_r in eq. (5).

2.1.2 *Electrostatic mirror force*

In addition to mechanical interactions, the lunar surface exhibits an inherent electric field, and its gravitational acceleration is approximately one-sixth that of Earth. In the interaction between lunar dust particles and the rover wheel, three types of non-contact physical forces are involved: electric field force, gravitational force, and electrostatic force. Under actual space conditions, when a positively charged lunar dust particle approaches the wheel surface, an equal but opposite charge is induced on the surface of the wheel, resulting in a Coulomb attraction between the particle and the wheel. Assuming the lunar dust particles are spherical and electrically insulating, the induced charge distribution on the particle surface becomes non-uniform due to the influence of the wheel's induced charge. This results in the concept of an equivalent charge position. According to prior studies, the radius of a lunar dust particle and the equivalent charge distance h are approximately linearly related. Specifically, h is about half of the curvature radius at the particle's apex. When the particle is approximated as a perfect sphere, this corresponds to one-quarter of the particle radius. Accordingly, a value of $h = R/4$ is adopted as the equivalent charge distance in this study.

Eqs. (10) and (11) indicate that the electrostatic attraction Fes is determined by the particle charge and the effective separation between the particle charge and the induced image charge on the wheel surface. Since the particle charge depends on the particle radius R and the electric potential ϕ_d, while the equivalent charge distance h is also related to particle size, the particle radius affects the electrostatic attraction through its effects on both the charge magnitude and the effective charge separation.

$$F_{es} = \frac{q^2}{4\pi\varepsilon_0(2h+2z_0)^2}, \tag{10}$$

$$q = 4\pi\varepsilon_0 R\phi_d, \tag{11}$$

where q is the charge of the moon dust particles; the vacuum capacitance $\varepsilon_0 = 8.8541 \times 10^{-12}$ F/m; the equivalent charge position $h = R/4$; the equilibrium spacing of the contact between the moon dust particles and the surface $z_0 = 0.4 \times 10^{-9}$ m; the potential of the near-moon surface $\phi_d = 4.1$ V.

2.1.3 *Electrostatic double-layer force*

When lunar dust particles come into contact with a surface, electron transfer occurs between materials until electrostatic equilibrium is reached. This process leads to a microscopic contact potential difference, generating an electrostatic double-layer force, calculated as

$$F_{el} = \frac{\pi\varepsilon_0 R U_e^2}{R+z_0}, \tag{12}$$

where the microscopic contact potential difference $U_e = 0.4$ V.

2.1.4 *Electric field force*

Due to the presence of a non-uniform electric field near the lunar surface, dust particles experience an electric field force given by

$$F_e = Eq, \tag{13}$$

where E is the strength of the lunar electric field and q is the charge of the lunar dust particles.

2.1.5 *Centrifugal force*

Dust particles adhering to the wheel rotate with it, experiencing centrifugal force, which is calculated by

$$F_c = \frac{m v_{\text{wheel}}^2}{R_{\text{wheel}}}, \tag{14}$$

$$m = \frac{4}{3}\rho\pi R^3, \tag{15}$$

$$v_{\text{wheel}} = \frac{v}{1-s}, \tag{16}$$

where m is the mass of a lunar dust particle; ρ is the density of the lunar dust particle, taken as 2900 kg/m^3; v_{wheel} is the actual rotational speed of the wheel; v is the traveling speed of the rover; and s is the slip rate.

2.2 Mechanical modeling of lunar dust particle desorption from wheels

In this study, surface roughness of the particles is taken into account, and the lunar dust particles are assumed to be spherical. As the particle volume fraction in the dust-plume region is extremely low (~10^{-6}–10^{-8}), the system falls well within the dilute regime defined in multiphase-flow literature, where particle-particle interactions can be safely neglected [26]. Although the wheel surface is considered rough, the influence of wheel tread patterns is ignored. This section focuses on the mechanical model of lunar dust particle migration toward the wheel and the model of their subsequent detachment from the wheel surface.

Let F_{v1} denote the van der Waals force between a lunar dust particle and the lunar surface, and F_{v2} denote the van der Waals force between the particle and the wheel surface. Let G represent the gravitational force acting on the particle, F_{es} represent the electrostatic mirror force, F_{el} represent the electrostatic double-layer force, and F_e represent the electric field force. The primary forces acting on lunar dust particles in the interaction between the lunar surface and the wheel are illustrated in Figure 2.

It has been shown that particles deposited on a surface can detach via three primary mechanisms: rolling detachment,

sliding detachment, and direct lift-off [27]. Under typical lunar surface conditions, there is no surrounding atmosphere, and thus no fluid-induced tangential drag force. Furthermore, assuming that the electric field acts only in the vertical direction, lunar dust particles are not subjected to tangential orientation forces. Therefore, in this study, the dominant mechanisms by which lunar dust particles detach from the surface and migrate toward the wheel are considered to be direct lift-off and rolling detachment.

(1) Rolling desorption

When the total torque at the circular contact edge between a lunar dust particle and the lunar surface becomes zero, the particle begins to rotate around that point and subsequently detaches from the surface, migrating toward the wheel. This condition can be expressed as

$$F_e R + F_{es} R + F_{el} R \geqslant F_v a_t + GR, \tag{17}$$

where R is the radius of the lunar dust particles. $a_t = \sqrt{C_r}\, a_{JKR}$ denotes the effective contact radius at the onset of separation (for particles resting on the lunar regolith surface, a_t corresponds to the JKR contact radius between a spherical particle and a flat regolith surface, corrected by the roughness factor as described in Section 2.1.1).

(2) Direct uplift

When the total normal detachment force acting on a lunar dust particle exceeds the force holding it to the surface, the particle is directly lifted off and migrates from the lunar surface to the wheel. The condition is given by

$$F_e + F_{es} + F_{el} \geqslant G + F_v. \tag{18}$$

If either of the above conditions is satisfied, the lunar dust particle detaches from the lunar surface and is transported by the rotating rover wheel.

For lunar dust particles adhering to the wheel surface, the dominant forces include gravitational force, van der Waals force, centrifugal force, electric field force, and frictional force. Assuming the wheel rotates clockwise, particles at the horizontal contact point ($\theta = 90°$ in polar coordinates) initially adhere to the wheel surface and are carried along by the rotation until detachment occurs. As previously noted, three detachment mechanisms are considered: rolling, sliding, and direct lift-off.

(1) Rolling desorption

As illustrated in Figure 3(a), when the total torque acting

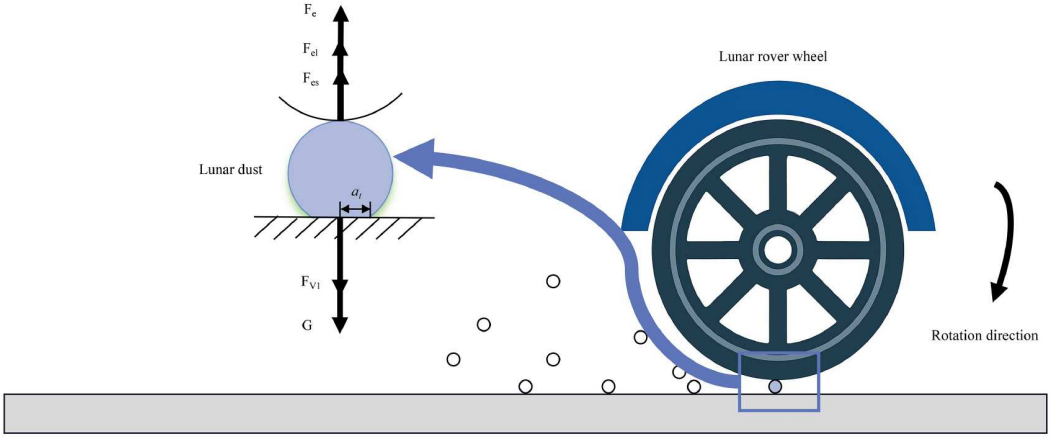


Figure 2 (Color online) Force analysis of moon dust particles between the moon surface-wheel surface.

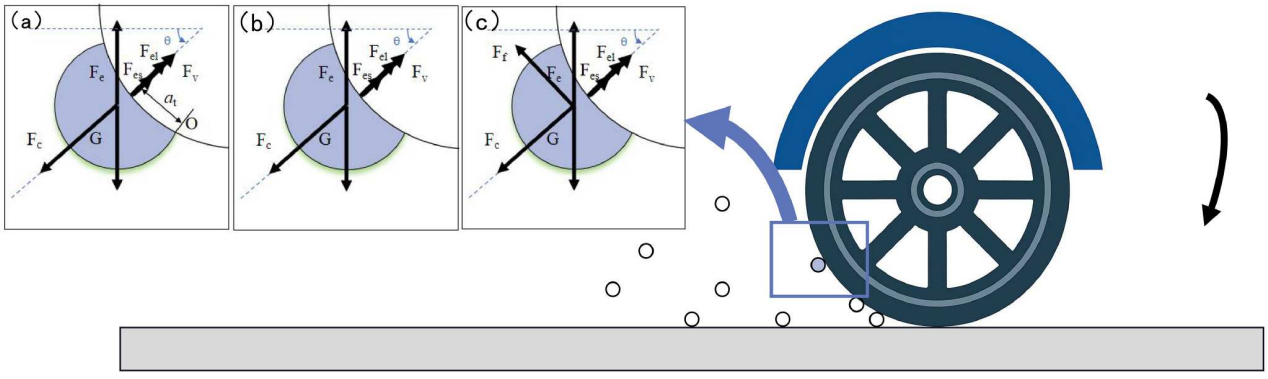


Figure 3 (Color online) Force analysis of lunar dust particles on smooth wheel surface. (a) Rolling desorption; (b) direct lift-off; (c) sliding desorption.

on a particle at the contact edge between the wheel surface and the particle becomes zero, rolling detachment occurs. The corresponding condition is

$$F_c a_t + \left|(G-F_e)\cos\theta\right| \sqrt{R^2 - a_t^2} - \left|(G-F_e)\right| \sin\theta a_t \geqslant (F_v + F_{es} + F_{el}) a_t, \tag{19}$$

where $a_t = \sqrt{C_r}\, a_{\mathrm{JKR}}$ is the contact radius considering roughness (for particles on the rover wheel surface, a_t incorporates the wheel-surface roughness and curvature effects).

(2) Direct lift-off

As illustrated in Figure 3 (b), direct lift-off occurs when the resultant normal force acting on a particle exceeds the total adhesion and gravity components:

$$F_c \geqslant F_v + F_{es} + F_{el} + \left|(G-F_e)\right| \sin\theta. \tag{20}$$

(3) Sliding desorption

As shown in Figure 3(c), when the tangential component of the resultant force exceeds the static frictional resistance, the particle undergoes sliding detachment from the wheel surface:

$$\left|(G-F_e)\cos\theta\right| \geqslant \mu \left| F_v + F_{es} + F_{el} + \left|(G-F_e)\right| \sin\theta - F_c \right|, \tag{21}$$

where $\mu = 0.3$ is adopted based on the reported static friction coefficient of lunar lander footpads [28].

Once any of the above detachment criteria are satisfied, the particle is released from the wheel surface and enters the dust lifting phase.

2.3 Simulation parameter configuration

The primary physical parameters of lunar dust include density, shear modulus, Poisson's ratio, and particle count.

The specific values used in the simulation are summarized in Table 1.

Based on the real design of the Apollo lunar rover wheels [29], the specific physical parameter settings of the wheels are shown in Table 2.

2.4 Particle size distribution and simulation strategy for lunar dust particles

2.4.1 Statistical characteristics of lunar dust particle size distribution

To enhance the realism of lunar dust particle inputs in the simulation program, this study adopts the statistical distribution of actual particle sizes obtained from lunar soil samples returned by the Chang'e 5 mission. These samples were collected using a drilling method, which allows access to subsurface lunar regolith and yields more representative particle data. According to analysis results of the Chang'e 5 lunar dust samples [30], 95% of the particles exhibit diameters ranging from 1.40 to 9.35 μm, with a mean particle size of 3.96 μm and a median of 2.90 μm. The corresponding particle size distribution is illustrated in Figure 4 and demonstrates a positively skewed log-normal distribution.

Let the diameter of the lunar dust particles be d. The lognormal distribution probability density function is shown in eq. (22):

$$f(d) = \frac{1}{d\sigma_3\sqrt{2\pi}} \exp\left(-\frac{(\ln d - \mu_1)^2}{2\sigma_3^2}\right), \tag{22}$$

where μ_1 and σ_3 are the mean and standard deviation of the distribution on a logarithmic scale:

$$\mu_1 = \ln(d_{\mathrm{median}}) = \ln(2.90 \times 10^{-6}) \approx 1.065,$$

Table 1 Lunar dust physical parameter settings

Parameter name	Value
Intensity ρ (kg/m^3)	2900
Modulus of elasticity E_1 (MPa)	65
Poisson's ratio μ_1	0.33
Gravitational acceleration of the lunar surface g_{lunar} (m/s^2)	1.6
Surface energy γ_1 (J/m^2)	0.05
Surface roughness σ_1 (μm)	0.1

Table 2 Wheel physical parameter settings

Parameter name	Value
Material	Aluminum alloy
Diameter D_0 (mm)	818
Wheel width W (mm)	229
Poisson's ratio μ_2	0.2
Modulus of elasticity E_2 (GPa)	70
Surface energy γ_2 (J/m^2)	0.9
Diameter of the outer side of the dust baffle D_2 (mm)	1000
Diameter of the inner side of the dust baffle D_1 (mm)	880
Surface roughness σ_2 (μm)	1

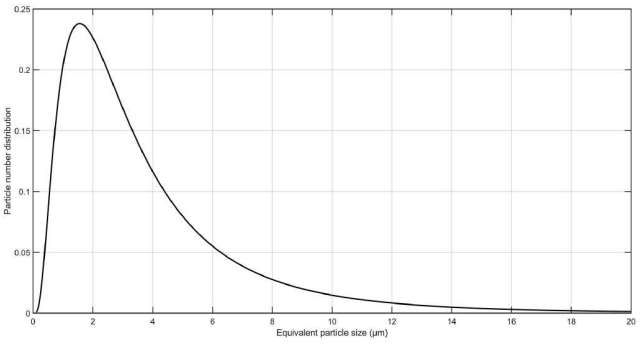


Figure 4 Quantitative distribution of particle sizes of Chang'e 5 lunar dust samples.

$$\sigma_3 = \sqrt{2\ln\frac{d_{\text{mean}}}{d_{\text{median}}}} = \sqrt{2\ln\frac{3.96}{2.90}} \approx 0.606.$$

2.4.2 *Kinetic dominance of large lunar dust particles*

Although the lunar soil samples returned from the Chang'e 5 mission show that micron-scale lunar dust particles are numerically dominated by small-sized particles, large-sized particles tend to dominate in dynamic behavior and dust transport modeling. Based on the computational results from the theoretical model described earlier, lunar dust particles larger than 45.3 μm do not adhere to the wheel and instead rotate with it. Furthermore, under a rover speed of 2.5 m/s and a slip rate of 0.2, particles smaller than 18.17 μm remain attached to the wheel surface and do not detach, thus initiating dust lifting.

This behavior is primarily attributed to the greater mass and higher inertia of large particles, which enables them to more readily overcome adhesive forces such as van der Waals and electrostatic forces, detach from the wheel surface, and be thrown off. Prior studies have also employed larger particles (e.g., 50 μm) as representative cases in dust modeling, owing to their dominant role in desorption, flight, and settling processes [4]. In practical missions, large particles are more likely to produce visible dust plumes under wheel-induced disturbances, thereby posing a greater risk of contaminating optical instruments and thermal control components.

2.4.3 *Method for generating lunar dust particle radii in this study*

Given the extremely small proportion of large-sized particles in the lognormal particle size distribution and the relatively flat probability density within that range, a uniform random sampling method is adopted in this study for generating particle radii during dust trajectory simulations. This approach improves computational efficiency while focusing on the kinetic behavior of particles that are likely to detach. It does not compromise the representativeness of dominant dust-raising particles but significantly reduces redundant calculations involving non-detachable particles, thereby enhancing the accuracy of dust lifting simulations.

The dust-lifting behavior of lunar rover wheels during traversal is simulated using the proposed theoretical model, with physical parameters defined in Tables 1 and 2 and a radius generation method as described above. Referring to the Apollo lunar rover wheel design, four values of traveling speed and four values of slip rate are selected using a scatter distribution function, as shown in Table 3.

The simulation time step is set to 0.01 s for each traveling speed and slip rate. At each time step, five lunar dust particles are generated based on the physical parameters mentioned above. The detachment angle for each particle is determined according to eqs. (19)–(21), and their trajectories following detachment from the lunar rover wheel are subsequently obtained. It should be noted that the use of uniform sampling affects only the visual density distribution of particles in the simulated plume. Since the objective of this study is to analyze detachment thresholds and trajectory dynamics rather than to reproduce the true statistical concentration of all particle sizes, this simplification does not influence the physical trends of interest. A log-normal radius distribution would produce a denser clustering of sub-10 μm particles but would not alter the calculated detachment conditions or the resulting particle kinematics.

2.5 Post-detachment trajectory dynamics of lunar dust particles

In the absence of an atmosphere on the lunar surface, the post-detachment motion of a lunar dust particle can be modeled as a ballistic trajectory governed primarily by the combined effects of gravity and the residual electric field. Accordingly, the horizontal (x) and vertical (z) velocity components of an individual particle at the moment of detachment can be formulated as shown in eqs. (23) and (24).

$$V_x = V\sin\theta, \tag{23}$$

$$V_z = V\cos\theta, \tag{24}$$

where V_x denotes the horizontal velocity (m/s) of an individual lunar dust particle at the moment of detachment and V_z denotes the corresponding vertical velocity (m/s); V represents the linear operating velocity (m/s) of the rover wheel, and θ is the particle detachment angle.

The horizontal and vertical positions of a dust particle after detachment from the wheel are obtained from eqs. (25) and (26).

$$X_0 = -R\cos\theta, \tag{25}$$

$$Z_0 = R - R\sin|\theta|, \tag{26}$$

where X_0 denotes the horizontal position (m) of an individual lunar dust particle at the moment of detachment, and Z_0 denotes its corresponding vertical position (m). R represents the wheel radius (m), and θ is the particle detachment angle.

Within a certain region above the lunar surface (i.e., the

Table 3 Simulated physical parameters of lunar dust and wheel

Variant	Parameter name	Distribution function	Range of values	Unit
v	Lunar rover travel speed	Scatter distribution	0.5, 1.0, 1.5, 2.0	m/s
s	Slip rate		0, 0.2, 0.4, 0.6	–

Debye layer), a non-uniform electric field exists and varies with height. For simplification, the electric field strength is assumed to be constant within this layer, whose height is taken as 0.41 m, and the dust particles are considered positively charged. After detachment from the wheel surface, the particle motion exhibits two possible scenarios. (1) Non-escape scenario: the particle cannot escape the Debye layer. Its upward velocity decreases until the deceleration becomes zero at some height within the layer, after which it accelerates downward and returns to the surface. (2) Escape scenario: the particle escapes the Debye layer. Once outside the layer, the electric field force vanishes and gravitational acceleration dominates. The particle then reaches a maximum height where its deceleration becomes zero, and subsequently descends back through the Debye layer and returns to the surface.

The net acceleration of a lunar dust particle within the Debye layer is obtained from eq. (27).

$$a_1 = \frac{G - F_e}{m}, \tag{27}$$

where a_1 is the net acceleration of the lunar dust particle within the Debye layer (m/s^2), G is the gravitational force acting on the particle (N), F_e is the electric field force exerted on the particle (N), and m is the particle mass (kg).

(1) Non-escape scenario

As illustrated in Figure 5, under these conditions, the maximum lofting height, the time required to reach this height, and the time for the particle to return to the surface can be calculated using eqs. (28)–(30).

$$H_{\max 1} = Z_0 + \frac{V_z^2}{2a_1}, \tag{28}$$

$$t_1 = \frac{V_z}{a_1}, \tag{29}$$

$$t_2 = \sqrt{\frac{2 \times 0.41}{a_1}} + t_1, \tag{30}$$

where $H_{\max 1}$ denotes the maximum lift-off height of the dust particle (m), t_1 represents the time required to reach this height (s), and t_2 is the time required for the particle to return to the lunar surface (s). a_1 denotes the net acceleration of the particle within the Debye layer (m/s^2), and V_z represents the initial vertical velocity of the particle at the moment of detachment (m/s).

Accordingly, the governing equation of motion for the lunar dust particle under these conditions can be expressed as

$$x = V_x \times t, \tag{31}$$

$$z = \begin{cases} Z_0 + V_z \times t - \frac{1}{2} \times a_1 \times t^2, \ (t \leqslant t_1), \\ H_{\max 1} - \frac{1}{2} \times a_1 \times (t - t_1)^2, \ (t_1 < t < t_2), \\ 0, \ (t \geqslant t_2). \end{cases} \tag{32}$$

(2) Escape scenario

As shown in Figure 6, the maximum lofting height $H_{\max 2}$, the time at which the particle leaves the Debye layer t_3, the time at which it re-enters the layer t_4, and the time required to reach the lunar surface t_5 can be calculated using eqs. (33)–(36).

$$H_{\max 2} = 0.41 + \frac{V_z^2 - 2 \times a_1 \times 0.41}{2 \times g_l}, \tag{33}$$

$$t_3 = \frac{\sqrt{\left(V_z^2 - 2 \times a_1 \times 0.41\right)}}{a_1}, \tag{34}$$

$$t_4 = t_3 + \frac{2 \times \sqrt{\left(V_z^2 - 2 \times a_1 \times 0.41\right)}}{g_l}, \tag{35}$$

$$t_5 = t_4 + \frac{2 \times \sqrt{\left(V_z^2 - 2 \times a_1 \times 0.41\right)}}{g_l} + \frac{V_z}{a_1}. \tag{36}$$

Accordingly, the governing equation of motion for the lunar dust particle under these conditions can be expressed as

$$x = V_x \times t, \tag{37}$$

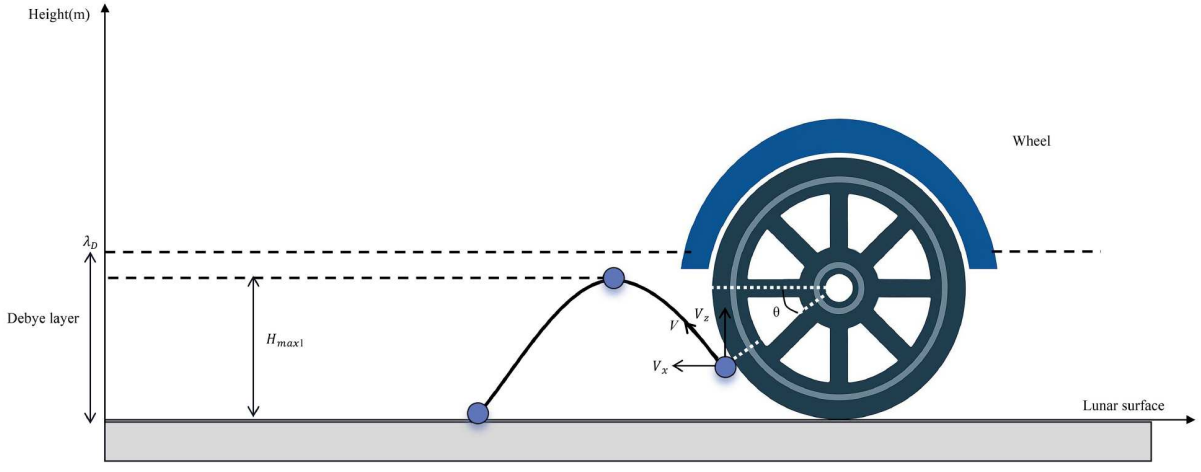


Figure 5 (Color online) Trajectory of a lunar dust particle that fails to escape the Debye layer.

$$z=\begin{cases} Z_0+V_z\times t-\dfrac{1}{2}\times a_1\times t^2, (t\leqslant t_3),\\ 0.41+\sqrt{\left(V_z^2-2\times a_1\times 0.41\right)}\times(t-t_3)\\ \quad -\dfrac{1}{2}\times g_l\times(t-t_1)^2, (t_3<t<t_4),\\ 0.41-\sqrt{\left(V_z^2-2\times a_1\times 0.41\right)}\times(t-t_4)\\ \quad -\dfrac{1}{2}\times a_1\times(t-t_4)^2, (t_4\leqslant t<t_5),\\ 0, (t\geqslant t_5). \end{cases} \tag{38}$$

Figure 7 presents the overall modeling framework of this study, summarizing the micro-scale force analysis, detachment criterion, initial angle/velocity determination, and particle trajectory simulation.

3 Results and discussion

3.1 Lunar dust desorption particle size and angle versus different parameters

Based on the established theoretical model, the effects of

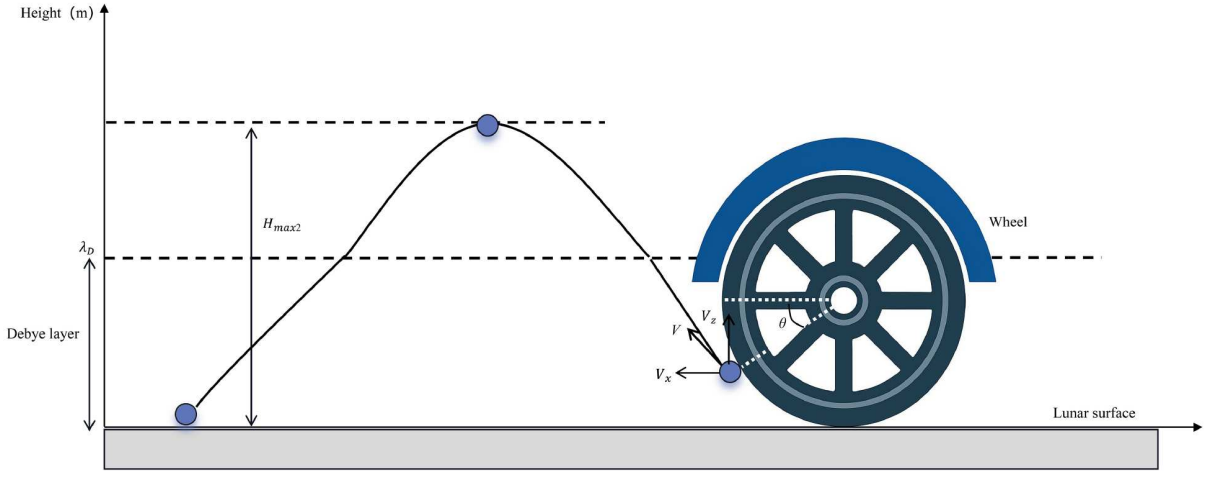


Figure 6 (Color online) Trajectory of a lunar dust particle that escapes the Debye layer.

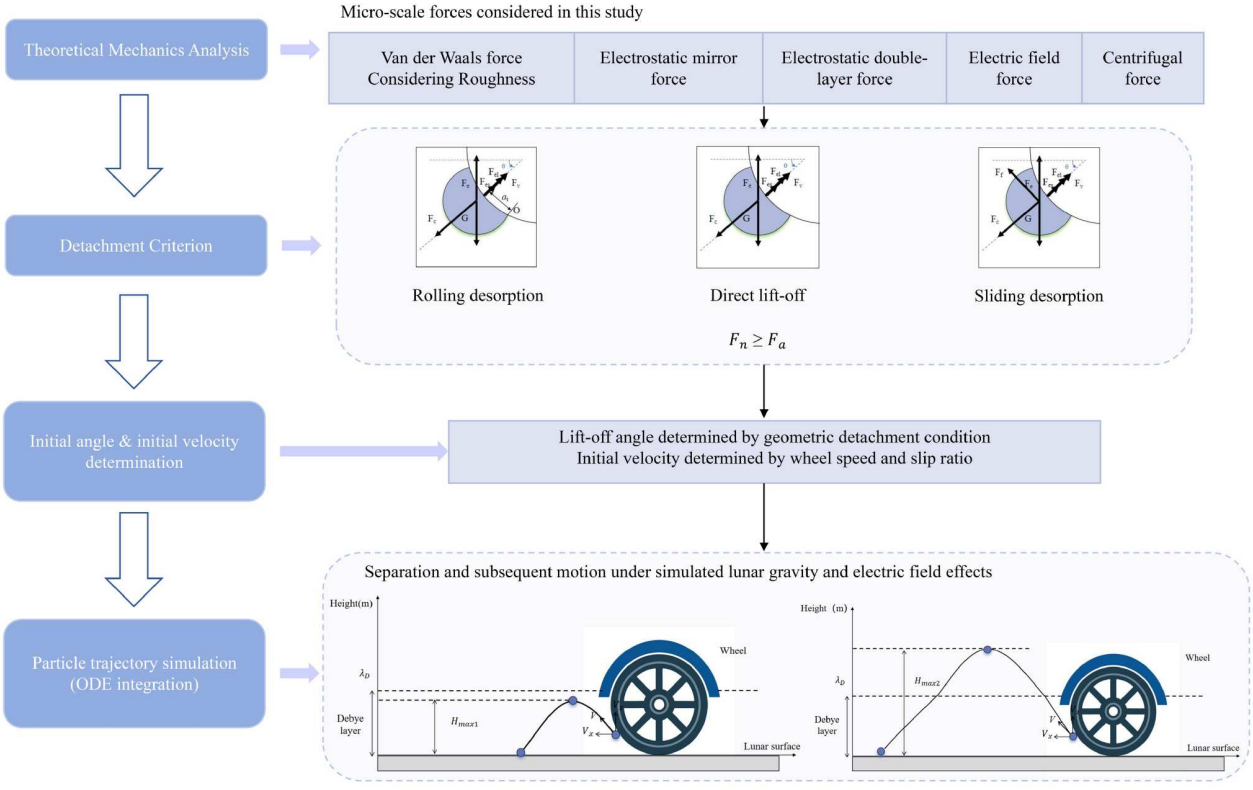


Figure 7 (Color online) Overall modeling framework of the proposed lunar dust detachment and trajectory simulation model.

lunar rover traveling speed and wheel slip ratio on the detachment radius and detachment angle of lunar dust particles were investigated. Under a fixed slip ratio of 0.2, Figure 8 presents the variation in these parameters with respect to different traveling speeds.

As illustrated in Figure 8, increasing the rover speed results in a decrease in the minimum detachment radius, indicating that dust particles tend to detach from positions closer to the wheel hub at higher speeds. This trend is attributed to the enhancement of centrifugal force, which scales with the square of wheel speed (as described in eq. (16)). The corresponding increase in centrifugal torque facilitates the satisfaction of the rolling detachment criterion (as described in eq. (17) and Figure 3(a)), enabling particles to detach from earlier positions on the wheel. In addition, the detachment angle shows an inverse relationship with particle radius: larger particles tend to detach at smaller angles. This is because the centrifugal torque increases substantially with particle size, while the adhesive torque, primarily governed by van der Waals and electrostatic forces, exhibits a relatively weak dependence on particle size. Consequently, larger particles overcome adhesion more easily and detach earlier in the wheel's rotation, whereas smaller particles require larger angular displacement to satisfy the detachment condition.

At a constant traveling speed of 1.5 m/s, the effect of slip ratio on the detachment radius and detachment angle of lunar dust particles was further analyzed. As the slip ratio increases, the effective rotational speed of the wheel also increases, thereby enhancing the centrifugal force acting on the particles (as described in eq. (16)). As shown in Figure 9, higher slip ratios result in a decrease in the minimum detachment radius, indicating that particles detach from positions closer to the wheel center under high-slip conditions. Mechanically, this trend arises from the fact that elevated slip ratios not only increase the normal detachment force but also enhance the tangential shear at the wheel-particle interface. This promotes a transition in the dominant detachment mechanism from rolling detachment, governed by torque balance (as described in eq. (17)), to sliding detachment, which occurs when the tangential component of the net force exceeds the frictional resistance (as described in eq. (19) and illustrated in Figure 3(c)). Moreover, as the slip ratio increases, the variation ranges of both detachment angle and detachment position radius become narrower. This is attributed to stronger centrifugal and shear forces diminishing the sensitivity of the detachment conditions to particle size, resulting in a more concentrated angular range for particle release, particularly for mid-sized particles. These simulation results highlight not only quantitative changes in detachment positions but also a qualitative shift in the detachment mechanism, reflecting the evolving dynamics of wheel-particle interactions under different slip conditions.

These findings are consistent with previous studies on

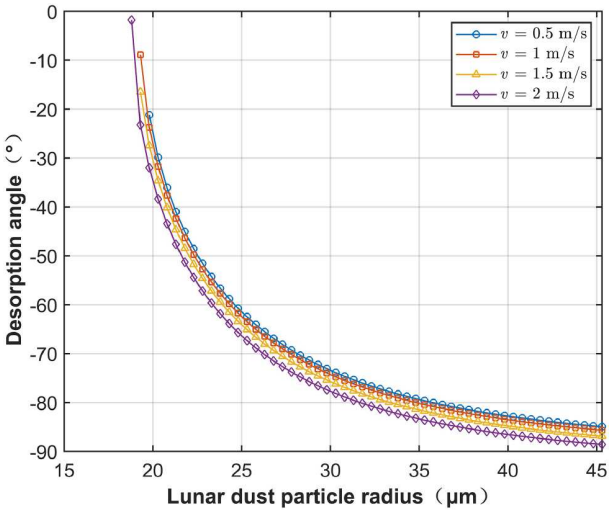


Figure 8 (Color online) Variation patterns of the desorption radius and angle of lunar dust particles at different lunar rover traveling speeds.

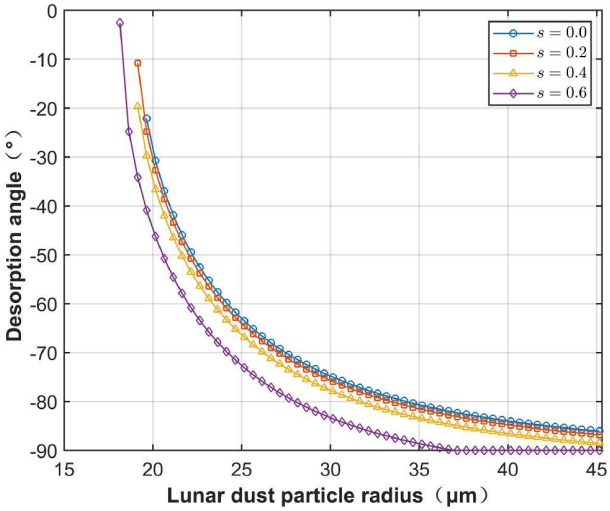


Figure 9 (Color online) Changing law of desorption radius and angle of lunar dust particles under different slip rates.

particle detachment mechanisms. Song et al. [24] observed in his investigation of ground vehicle wheel-dust interactions that, under comparable conditions, larger dust particles are more likely to detach than smaller ones, and higher wheel speeds promote particle detachment and suspension. It is important to note, however, that in the lunar vacuum environment, aerodynamic drag and wake-induced torque are negligible. As a result, the influence of lunar rover speed on dust detachment is governed solely by centrifugal force and gravity, without the additional aerodynamic detachment mechanisms present in terrestrial conditions. This explains the relatively moderate response of detachment trends to speed variations observed in the current simulations.

Overall, the results confirm the validity of the mechanism-based rolling detachment model and underscore its pre-

dominant influence under representative lunar surface conditions.

3.2 Impact of rover travel velocity and slip ratio on lunar dust dispersion

This study presents a systematic analysis of lunar dust dispersion behavior under varying lunar rover travel speeds (0.5–2.0 m/s) and slip ratios (0–0.6). Figures 10 and 11 depict the dust trajectories at the wheel cross-sectional plane, while Figures 12 and 13 quantitatively illustrate the relationships between dust height, horizontal travel distance, and the respective speed and slip parameters. The results indicate that at lower vehicle speeds or slip ratios, the detached particles exhibit limited flight distance and altitude, with dust primarily confined to the near-surface region behind the wheels. As the travel speed or slip ratio increases, the trajectories of the particles exhibit more pronounced parabolic characteristics, and the overall dust dispersion range expands significantly, reaching vertical heights that may exceed the height of the vehicle itself (0.818 m).

By integrating image-based observations with mechanical

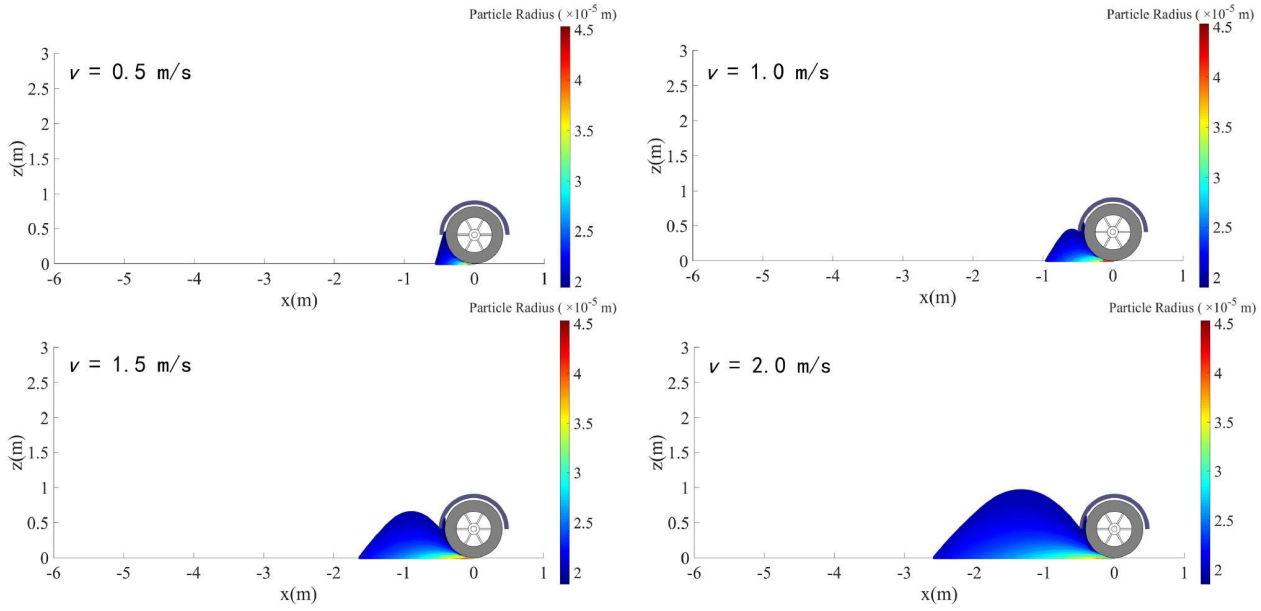


Figure 10 (Color online) Dust lifting patterns at lunar rover traveling speeds of 0.5–2.0 m/s.

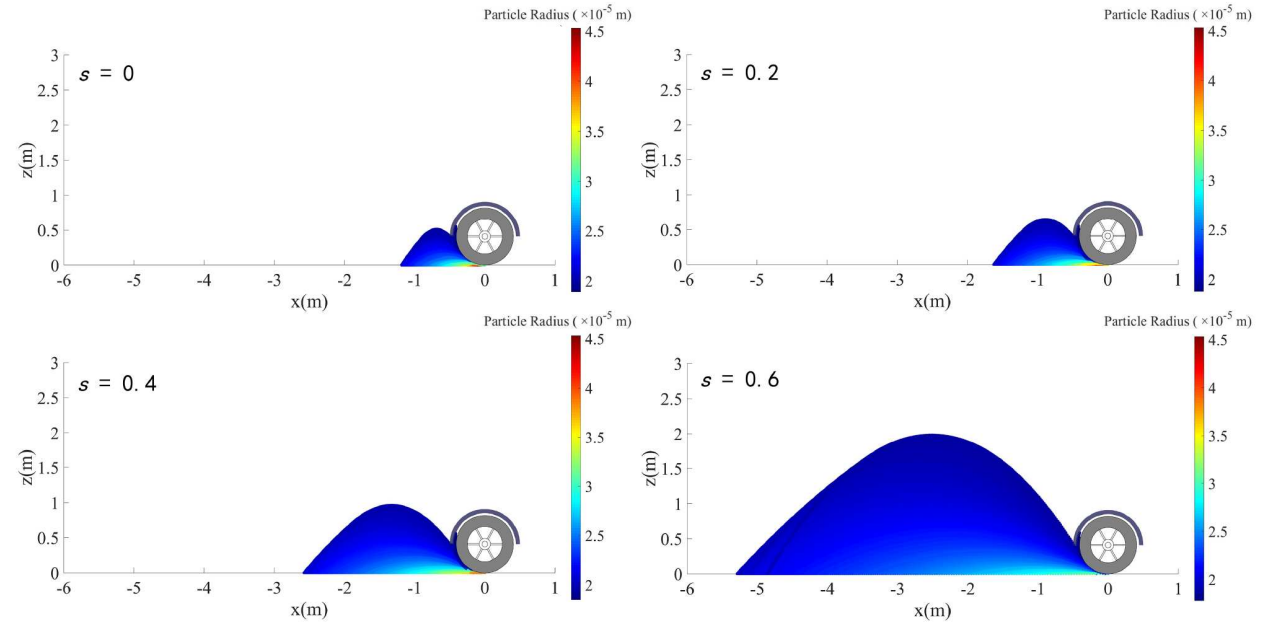


Figure 11 (Color online) Dust lifting patterns at lunar rover slip rates of 0–0.6.

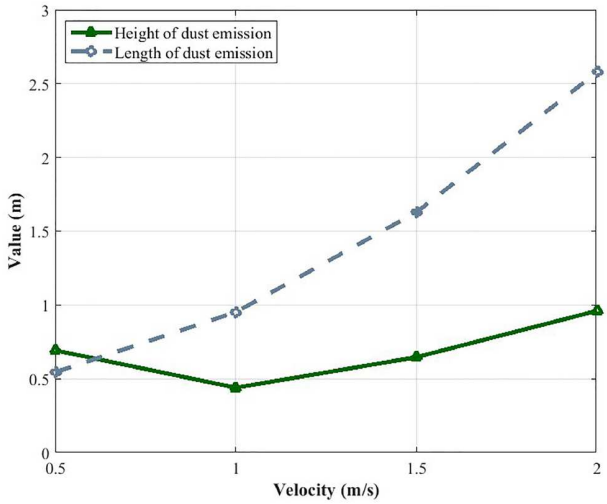


Figure 12 (Color online) Relationship between rover traveling speed and dust characterization parameters.

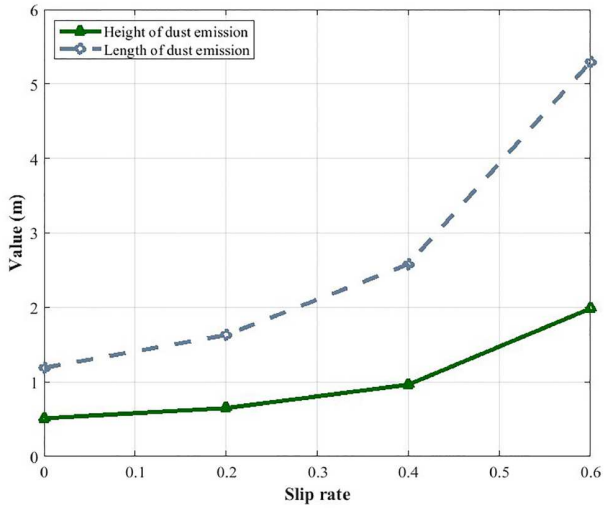


Figure 13 (Color online) Relationship between rover slip rate and dust characterization parameters.

model analysis, the detachment and flight behavior of lunar dust particles exhibit a clear particle size-dependent stratification. Larger particles, due to greater inertia and smaller detachment angles, tend to settle rapidly near the ground, forming a dense near-surface deposition zone. Medium-sized particles demonstrate the broadest spatial coverage in their trajectories. In contrast, smaller particles require larger detachment angles to overcome the adhesive torque, which is primarily induced by van der Waals and electrostatic forces. This leads to higher initial vertical velocities and contributes to the formation of a high-altitude dispersion zone. This zonal structure corresponds well with the rolling and sliding detachment mechanisms described in eqs. (17) and (19), particularly under high slip conditions.

An interesting phenomenon is observed when the vehicle speed increases from 0.5 to 1.0 m/s: the maximum dust height decreases. This effect is attributed to interception by the dust shield, which prevents high-altitude particles from dispersing further and underscores the importance of structural design considerations in the evaluation of dust behavior.

In alignment with previous research (e.g., Zhang et al. [4]), this study confirms that increased speed and slip ratio intensify dust dispersion. However, a distinguishing aspect of this work lies in the inclusion of particle-size-dependent modeling, which reveals a spatially stratified dust structure dominated by particle diameter. Therefore, this study reveals not only the particle-size-dependent stratification in dust dispersion but also provides a theoretical basis for understanding the underlying mechanisms of dust behavior under varying operational conditions. The engineering implications derived from these findings, including risk zones and mitigation strategies for lunar rover design, are further discussed in the following section.

3.3 Comparative analysis of simulation results and real dust images of lunar dust

Based on the aforementioned simulation method, the dust dispersion behavior of the lunar rover was simulated at various time points under a traveling speed of 1.8 m/s and a slip rate of 0.2. These results were qualitatively compared with a screenshot from the Apollo 16 lunar rover video, captured on April 21, 1972.

It should be noted that the comparison with the Apollo 16 rover imagery is necessarily qualitative. The available video data do not provide sufficiently accurate information on particle size distribution, initial velocity, detachment location, or local electric-field conditions, which precludes rigorous quantitative validation. In addition, the Apollo footage represents a three-dimensional dust plume projected onto a two-dimensional camera frame, whereas the present model simulates a two-dimensional cross-section of particle motion. This geometric mismatch, together with uncertainties in rover operating parameters (e.g., exact slip ratio, wheel torque, and soil conditions) and camera-related effects such as viewing angle, exposure, and motion blur, imposes additional limitations on direct numerical comparison.

Nevertheless, the simulated dust-lofting trends, dispersion patterns, and the variation in maximum dust height with rover speed shown in Figure 14 are broadly consistent with the observable behaviors captured in the Apollo mission imagery. Thus, the purpose of this comparison is to demonstrate the physical plausibility of the proposed mechanism rather than to achieve exact numerical agreement. More stringent quantitative verification would require high-resolution optical measurements or controlled lunar-regolith experiments, which fall outside the scope of the present study.

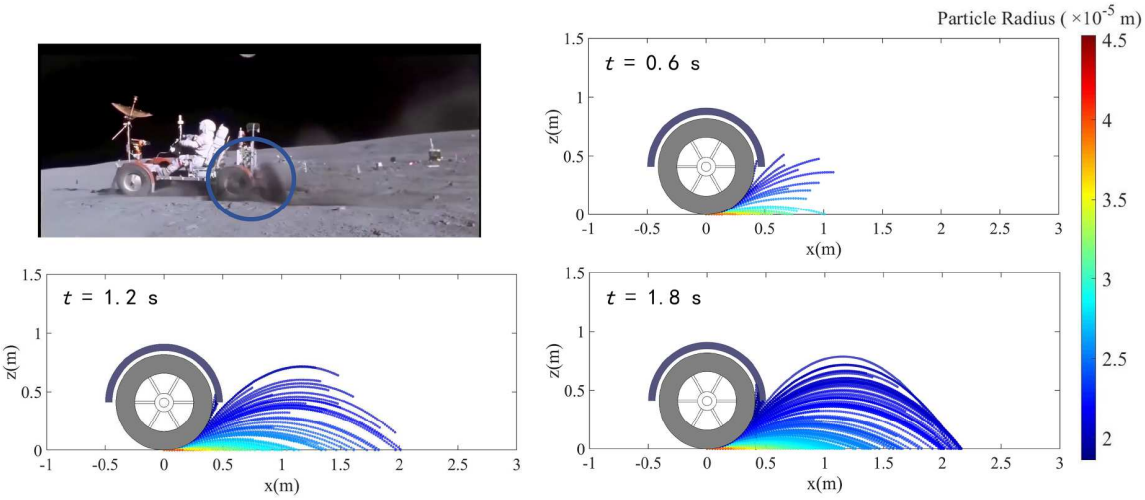


Figure 14 (Color online) Comparison of real cases and simulation results.

3.4 Engineering recommendations

Based on the simulation and theoretical analysis of lunar dust dispersion, several engineering recommendations can be proposed to enhance the reliability and dust resilience of lunar rover systems.

(1) Speed and slip control

Limit the operational speed of the rover to below 2.0 m/s and maintain the slip ratio around 0.2 to suppress excessive dust lifting and reduce the risk of interference with sensitive components.

(2) Zone-based protection strategy

(i) For the near-surface deposition zone (particle diameter > 60 μm), prioritize abrasion-resistant materials and anti-clogging design on wheel edges and lower structural parts.

(ii) For the mid-layer migration zone (particle diameter 40–60 μm), incorporate protective covers and dust-tolerant enclosures for side-mounted sensors.

(iii) For the high-altitude dispersion zone (particle diameter < 40 μm), reinforce the sealing and shielding of upper optical systems, antennas, and thermal radiators.

(3) Dust shield optimization

Adjust the geometry and positioning of dust shields to intercept particles most likely to reach high altitudes, especially at medium speeds (−1.0 m/s) where upward dust flux is significant.

These measures, derived from physics-based modeling and validated by image simulations, provide a technical foundation for mitigating lunar dust hazards and ensuring sustained functionality of exploration systems in the extreme lunar environment.

4 Conclusion

In this paper, the dusting behavior of lunar dust particles was simulated using a numerical model under varying traveling speeds and slip rates of the lunar rover. The simulation results indicate that with increasing traveling speed and slip rate, the dust-raising phenomenon becomes more pronounced, characterized by a continuous increase in both the maximum dusting height and maximum dusting distance. Under high-speed and high-slip conditions, the vertical height of lunar dust particles can reach up to 1.8 m, significantly exceeding the wheel body height (0.818 m), thus posing a potential threat to the structural integrity and mission performance of the rover. Further analysis reveals a clear particle size-dependent partitioning in dust behavior: large particles tend to accumulate in the near-surface deposition zone, medium-sized particles are distributed within a mid-level migration zone, while fine particles dominate the high-altitude dispersion zone. This particle size-governed spatial distribution not only elucidates the physical mechanism of lunar dust trajectories but also provides specific engineering guidance for the protection of critical rover components. For instance, near-surface particles may lead to abrasion and clogging of wheel rims and structural elements; mid-level dust may obscure sensor fields of view; and fine particles at high altitudes could contaminate optical lenses, communication antennas, and thermal control systems. Therefore, region-specific protection strategies should be formulated to optimize vehicle structural design and improve dust mitigation approaches.

To verify the validity and applicability of the proposed model, simulated dust images were compared with actual images obtained during the operation of the Apollo lunar rover. The results demonstrate a high degree of agreement in particle size distribution, trajectory structure, and vertical/horizontal dispersion characteristics, thereby confirming the model's accuracy in describing the underlying physical mechanisms and in parameter calibration. The observed differences in particle responses, including force interac-

tions, detachment behavior, and subsequent trajectories across various particle sizes, provide quantifiable insights into the fundamental nature of dusting and the key factors governing its control.

From a mechanical perspective, this study establishes a desorption model for lunar dust particles that incorporates van der Waals, electrostatic, and Coulomb forces. Coupled with operating parameters such as velocity and slip rate, the model enables systematic analysis of dusting conditions and distribution characteristics. These findings provide a theoretical foundation and technical reference for the dust-proof structural design of future lunar rovers, strategic equipment placement, and the successful execution of lunar surface missions.

Although the proposed model captures the essential microscale mechanisms governing lunar dust detachment and successfully reproduces key trends observed in Apollo mission imagery, several limitations remain due to the absence of quantitative, particle-scale measurements from actual rover operations. Future research should therefore focus on conducting controlled laboratory experiments with lunar-regolith simulants to obtain direct measurements of adhesion forces, detachment thresholds, surface roughness effects, and electric-field distributions under realistic conditions. Interparticle interactions, such as particle-particle collisions and clustering behavior, should also be incorporated into subsequent analyses to better represent the collective dynamics of dust-lifting events. Furthermore, extending the present two-dimensional framework into a fully three-dimensional model and applying realistic log-normal particle-size distributions will further improve the fidelity of predicted dust-plume morphology and particle concentration fields. Together, these advancements will contribute to a more comprehensive and experimentally validated foundation for dust-mitigation strategies and robust rover design in future lunar surface missions.

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References

1 Lin Y, Yang W, Zhang H, et al. Return to the Moon: New perspectives on lunar exploration. Sci Bull, 2024, 69: 2136–2148
2 Chen Y, Hu S, Li J-H, et al. Chang'e-5 lunar samples shed new light on the Moon. Innov Geosci, 2023, 1: 100014
3 Yang M F, Zhang G, Zhang W, et al. Technical design and implementation of Chang'e-5 robotic sample return mission on lunar surface (in Chinese). Sci Sin Tech, 2021, 51: 738–752
4 Zhang K, Wu J, Zhang Y, et al. Dust emission evaluation of the flexible metal wheel over lunar terrain and fender effect. Mech Based Des Struct Machines, 2024, 52: 7522–7547
5 Gaier J R. Effect of illumination angle on the performance of dusted thermal control surfaces in a simulated lunar environment. In: Proceedings of the 39th International Conference on Environmental Systems (ICES). Reston: AIAA, 2009
6 Walton OR. Adhesion of lunar dust. NASA Technical Report NASA-CR-2007-214685, 2007
7 Hartzell C M, Scheeres D J. The role of cohesive forces in particle launching on the Moon and asteroids. Planet Space Sci, 2011, 59: 1758–1768
8 Wang Z, Tian D, Ma Z, et al. An electrostatic lofting model based on adhesion separation characteristics for lunar dust. Planet Space Sci, 2020, 184: 104883
9 Berkebile S, Street K, Gaier J. Adhesion between volcanic glass and spacecraft materials in an airless body environment. In: Proceedings of the 3rd AIAA Atmospheric Space Environments Conference. Reston: AIAA, 2011
10 Yeo L H, Wang X, Dove A, et al. Laboratory investigations of triboelectric charging of dust by rover wheels. Adv Space Res, 2023, 72: 1861–1869
11 Guo Z. Study on dust lifting and dust adhesion-friction characteristics of lunar rover wheels (in Chinese). Dissertation for Master's Degree. Jilin: Jilin University, 2022
12 Stubbs T J, Vondrak R R, Farrell W M. A dynamic fountain model for lunar dust. Adv Space Res, 2006, 37: 59–66
13 Wang H, Phillips Iii J R, Dove A R, et al. Investigating particle-particle electrostatic effects on charged lunar dust transport via discrete element modeling. Adv Space Res, 2022, 70: 3231–3248
14 Mullis CH. A study and analysis of the MSFC lunar roving vehicle dust profile test program. NASA Technical Report NASA-CR-121075, 1971
15 Xia K. Finite element modeling of tire/terrain interaction: Application to predicting soil compaction and tire mobility. J Terra Mech, 2011, 48: 113–123
16 González Cueto O, Iglesias Coronel C E, Recarey Morfa C A, et al. Three dimensional finite element model of soil compaction caused by agricultural tire traffic. Comput Electron Agr, 2013, 99: 146–152
17 Nakashima H, Fujii H, Oida A, et al. Parametric analysis of lugged wheel performance for a lunar microrover by means of DEM. J Terra Mech, 2007, 44: 153–162
18 Zhu J, Zou M, Shen Y, et al. 3D DEM-FEM simulation of the flexible metal wheel-soil interaction in low gravity environments. Proc Institution Mech Engineers Part C-J Mech Eng Sci, 2023, 237: 1267–1278
19 Zhao L, Zhang R, Du Y, et al. Research in innovative mesh wheel with bionic wheel surfaces and lugs for superior lunar rover performance. Acta Astronaut, 2025, 228: 853–864
20 Wang H. Simulation and experimental study on lunar rover wheel-induced dust lifting (in Chinese). Dissertation for Master's Degree. Harbin: Harbin Institute of Technology, 2013
21 Johnson K L, Kendall K, Roberts A D. Surface energy and the contact of elastic solids. Proc R Soc London A Math Phys Sci, 1971, 324: 301–313
22 Derjaguin B V, Muller V M, Toporov Y P. Effect of contact deformations on the adhesion of particles. J Colloid Interface Sci, 1975, 53: 314–326
23 You S, Wan M P. Mathematical models for the van der Waals force and capillary force between a rough particle and surface. Langmuir, 2013, 29: 9104–9117
24 Song J, Qian H, Zheng X. Particle detachment due to wheel rotation. Indoor Built Environ, 2021, 30: 1312–1322
25 Cheng D. Mechanical Design Handbook (in Chinese). 6th ed. Beijing: Mechanical Industry Press, 2016
26 Gidaspow D. Multiphase Flow and Fluidization: Continuum and Kinetic Theory Descriptions. New York: Academic Press, 1994
27 Henry C, Minier J P. Progress in particle resuspension from rough surfaces by turbulent flows. Prog Energy Combust Sci, 2014, 45: 1–53
28 Kong L. Study on the dust lofting characteristics of charged lunar regolith (in Chinese). Dissertation for Master's Degree. Shandong: Shandong Jianzhu University, 2022
29 Asnani V, Delap D, Creager C. The development of wheels for the Lunar Roving Vehicle. J TerraMech, 2009, 46: 89–103
30 Li C, Hu H, Yang M F, et al. Characteristics of the lunar samples returned by the Chang'E-5 mission. Natl Sci Rev, 2022, 9: nwab188